

Welcome to img2avif

img2avif is a powerful command-line tool that converts your images to the modern AVIF format. Whether you're a professional photographer working with RAW files or just someone who wants to save disk space without sacrificing quality, img2avif gives you fine-grained control over the conversion process.

What is AVIF? AVIF (AV1 Image File Format) is a modern image format that typically produces files 50% smaller than JPEG while maintaining better visual quality. It supports high dynamic range, wide color gamuts, and transparency.

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Quick Start

Basic Usage

```
# Convert a single image
img2avif -i photo.jpg

# Convert all images in a directory
img2avif -d ./my-photos

# Convert with custom quality
img2avif -i photo.jpg -q 85

# Process a directory recursively
img2avif -r ./photos --delete
```

What Happens

When you run img2avif, the tool:

1. Reads your input image (supports over 30 formats including JPEG, PNG, and RAW)
 2. Applies any requested adjustments (exposure, rotation, etc.)
 3. Encodes the result as an AVIF file
 4. Saves it alongside your original (or replaces it if you choose)
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Basic Photography Concepts

Before diving into parameters, let's understand some fundamental concepts that `img2avif` uses.

Exposure

What it is: Exposure controls how bright or dark your image appears. It's like adjusting the amount of light that reaches a camera sensor.

In `img2avif`: For RAW files, you can adjust exposure compensation (0.5× to 5×). The `auto` setting analyzes your image's histogram to find optimal brightness.

Example: An underexposed (too dark) night photo can be brightened using `--raw-exposure 1.5`.

Contrast

What it is: Contrast defines the difference between the darkest and brightest parts of your image. High contrast makes images "pop" but can lose detail in shadows and highlights. Low contrast looks softer and more muted.

In `img2avif`: The `--raw-contrast` parameter (0.5 to 2.0) adjusts this. Values below 1.0 reduce contrast, above 1.0 increase it.

Saturation

What it is: Saturation controls the intensity of colors. Desaturated images appear muted or black-and-white. Oversaturated images look vivid but unnatural.

In `img2avif`: The `--raw-saturation` parameter (0.5 to 2.0) adjusts color intensity. Auto-saturation intelligently compensates when you increase exposure.

Bit Depth

What it is: Bit depth determines how many distinct colors or brightness levels can be represented per channel (Red, Green, Blue).

Bit Depth	Levels per Channel	Best For
8-bit	256	Standard displays, web images
10-bit	1,024	Smooth gradients, HDR
12-bit	4,096	Professional photography

In `img2avif`: Use `--bit-depth` to choose. Auto mode selects appropriate depth based on your source image.

Color Space: YUV vs RGB

RGB: Stores images using Red, Green, and Blue channels. Excellent quality but larger file sizes.

YUV: Separates brightness (Y) from color information (U and V). Allows compressing color data more aggressively since human eyes are less sensitive to color detail.

Color Space	Quality	File Size	Best For
RGB	Maximum	Largest	Graphics with text, sharp edges
YUV 4:4:4	Excellent	Medium	Photos with fine color detail
YUV 4:2:0	Good	Smallest	Most photos

Lossless vs Lossy Compression

Lossy (default): Removes some visual information to achieve smaller files. Quality settings from 0-100 let you trade off between size and perfection.

Lossless: Preserves every single pixel exactly as in the original. Creates larger files but mathematically perfect reproduction.

Command Line Interface

Basic Syntax

```
img2avif [OPTIONS] --input <FILE> | --directory <DIR> | --recursive <DIR>
```

Input Selection (Choose One)

Option	Description
-i, --input <FILE>	Convert a single file
-d, --directory <DIR>	Convert all images in a directory (non-recursive)
-r, --recursive <DIR>	Convert all images in directory and all subdirectories

Output Control

Option	Description
-o, --output <PATH>	Output file or directory. If converting a directory, this should be a folder
--delete	Delete original files after successful conversion
--discard-if-larger	Don't save AVIF if it's larger than original

Quality Settings

Option	Description	Default
-q, --quality <0-100>	Main compression quality	80
--alpha-quality <0-100>	Quality for transparent areas	80
-s, --speed <0-10>	Encoding speed (0=slowest/best compression)	4
--lossless	Enable lossless mode (ignores quality setting)	false

Color Configuration

Option	Description	Default
--color-space <yuv420 yuv444 rgb>	Color encoding method	yuv420
--bit-depth <8 10 12 auto>	Bits per color channel	auto

RAW Processing

Option	Description	Default
--raw-exposure <auto 0.5-5.0>	Exposure compensation	auto
--raw-percentile <0.5-1.0>	Target brightness percentile for auto-exposure	0.95
--raw-saturation <auto 0.5-2.0>	Color intensity adjustment	auto
--raw-contrast <0.5-2.0>	Contrast adjustment	1.0

Rotation

Option	Description	Default
--rotate <auto none 90 180 270>	Image rotation	auto

Metadata and Output

Option	Description	Default
--keep-metadata	Preserve EXIF metadata	false
-q, --quiet	Suppress all messages	false
-v, --verbose	Show detailed processing information	false

Note: Short options -q appears twice: for quality and quiet mode. They are distinguished by context - -q 85 sets quality, -q alone (no value) enables quiet mode.

Configuration Parameters (Detailed)

Quality (-q, --quality)

Range: 0 to 100

Default: 80

Controls the trade-off between file size and visual quality.

Quality	Typical Use	File Size (relative)
100	Maximum quality, almost lossless	~80% of original
90	Excellent quality for archiving	~40% of original
80	Good quality, great for web	~25% of original
70	Acceptable quality, small files	~15% of original
50	Low quality, tiny files	~8% of original

Tip: Unlike JPEG, AVIF at quality 80 typically looks indistinguishable from the original while being much smaller.

Speed (-s, --speed)

Range: 0 to 10

Default: 4

Controls encoding time vs compression efficiency.

Speed	Encoding Time	File Size at Same Quality
0-1	Very slow (10× slower)	10-20% smaller
2-3	Slow	5-10% smaller
4-5	Balanced (recommended)	Baseline
6-8	Fast	5-15% larger
9-10	Very fast	20-30% larger

Recommendation: Start with 4. Use slower speeds for archiving, faster speeds for previews or when converting many images.

Color Space (--color-space)

Yuv420 (default)

- **Best for:** Most photographs
- **Size saving:** Up to 50% vs RGB
- **Quality impact:** Minimal for natural images

Yuv444

- **Best for:** Images with fine colored text, computer graphics
- **Size saving:** Moderate (15-25% vs RGB)
- **Quality impact:** Preserves full color resolution

RGB

- **Best for:** Images requiring perfect color reproduction
- **Size saving:** Minimal
- **Quality impact:** None (mathematically lossless within quality constraints)

Bit Depth (--bit-depth)

Auto mode analyzes your source image:

- 8-bit for standard JPEGs and PNGs
- 10+ bit for HDR or high-bit-depth RAW files

Manual selection:

- **8-bit:** Universally compatible, smallest files
- **10-bit:** Better gradients (reduces banding in skies)
- **12-bit:** Maximum quality for professional work

Discard If Larger (- -discard-if-larger)

Prevents AVIF files that are larger than the original. Some images (especially already-compressed JPEGs with high quality settings) may not compress further.

Example output when enabled:

Skipping: AVIF larger than original (145.2 KB > 142.1 KB)

Delete Original (- -delete)

Warning: This permanently removes original files after successful conversion. Only use this when you have verified the results.

The tool never deletes files if conversion fails.

RAW Processing

img2avif supports over 50 RAW formats including:

- **Nikon:** NEF, NRW
- **Canon:** CR2, CR3, CRW
- **Sony:** ARW, SRF, SR2
- **Fujifilm:** RAF
- **Olympus:** ORF
- **Panasonic:** RW2
- **Pentax:** PEF
- **Sigma:** X3F
- **Adobe:** DNG
- And many more...

How RAW Processing Works

RAW files contain unprocessed sensor data. img2avif performs several steps:

1. **Demosaicing:** Converts the Bayer pattern (colored pixels) into full RGB
2. **White balance:** Applies camera's recorded color balance
3. **Exposure adjustment:** Brightens or darkens the image
4. **Saturation & contrast:** Applies your adjustments
5. **Gamma correction:** Converts to viewable colors
6. **AVIF encoding:** Compresses the final result

Auto-Exposure (**--raw-exposure auto**)

The auto-exposure algorithm:

1. Samples pixels throughout the image
2. Builds a brightness histogram
3. Finds the **--raw-percentile** brightness level (default 95th percentile)
4. Adjusts exposure so this level becomes 95% of maximum brightness

Example: At 95th percentile, the brightest 5% of pixels (specular highlights, light sources) are allowed to clip, while preserving detail in the rest of the image.

Auto-Saturation

When you increase exposure, colors can appear washed out. Auto-saturation compensates by boosting saturation by 20% of the exposure increase:

$$\text{auto_saturation} = 1.0 + (\text{exposure} - 1.0) \times 0.2$$

This maintains vibrant colors when brightening dark images.

Percentile Targeting (**--raw-percentile**)

Percentile	Effect
0.95 (95%)	Standard, preserves most highlights
0.99 (99%)	Very conservative, protects almost all highlights
0.85 (85%)	Brighter image, may clip more highlights

Use case: Night photos with bright lights benefit from lower percentiles (0.90-0.95). Scanned slides might need higher percentiles (0.98-0.99).

Manual Exposure Example

```
# Brighten an underexposed image by 1.5 stops
img2avif -i dark_raw.nef --raw-exposure 1.5

# Darken an overexposed image
img2avif -i bright_raw.cr2 --raw-exposure 0.7
```

Image Rotation

img2avif handles rotation in three ways.

Auto Rotation (**--rotate auto**) - Default

Reads orientation information from EXIF metadata (the "Orientation" tag that cameras write when you take a portrait photo). The tool automatically rotates the image to view correctly.

Supported orientations: Normal, 90°, 180°, 270°, mirrored variations.

Manual Rotation (**- -rotate 90, 180, 270**)

Forces rotation regardless of EXIF. Useful when:

- EXIF data is missing or incorrect
- You want to rotate images that don't have orientation tags
- Batch processing images that all need the same rotation

No Rotation (**- -rotate none**)

Disables all rotation. The image stays exactly as decoded.

Rotation and RAW Files

RAW files often store orientation information too. `img2avif` reads this via `rawloader` and applies it when `- -rotate auto` is specified.

Batch Processing

Directory Mode (**- d**)

Processes all images in a single directory (non-recursive):

```
img2avif -d ./vacation-photos -o ./converted
```

Output structure:

vacation-photos/		converted/
├─ beach.jpg	→	├─ beach.avif
├─ sunset.png	→	├─ sunset.avif
└─ family.tiff	→	└─ family.avif

Recursive Mode (**- r**)

Processes all images in a directory and every subdirectory:

```
img2avif -r ./photo-archive
```


Output structure preserves hierarchy:

photo-archive/ ├── 2023/ │ ├── spring/ │ │ └── flowers.jpg → │ ├── summer/ │ │ └── beach.jpg → └── old/ └── scanned.png →	photo-archive/ ├── 2023/ │ ├── spring/ │ │ └── flowers.avif │ ├── summer/ │ │ └── beach.avif └── old/ └── scanned.avif
--	---

Progress Display

When processing multiple files, img2avif shows:

- A progress bar with current file / total count
- Estimated time remaining
- Summary at the end (successful, failed, total size)

Verbose Mode for Debugging

```
img2avif -d ./photos -v
```

Shows:

- Each file being loaded
 - RAW metadata (camera model, white balance, black levels)
 - Auto-exposure calculation
 - Color mode detection
 - Rotation applied
 - Final compression details
-

Examples

Example 1: Simple Conversion

```
img2avif -i my-photo.jpg
```

Creates `my-photo.avif` in the same folder with default settings (quality 80, YUV420, speed 4).

Example 2: High-Quality Archive

```
img2avif -i wedding.nef -q 95 --color-space rgb --bit-depth 10 -s 2
```

- RAW file converted with very high quality
- RGB color space for maximum fidelity
- 10-bit depth for smooth gradients
- Slower encoding for better compression

Example 3: Web-Ready Images

```
img2avif -d ./originals -o ./web --quality 75 --discard-if-larger
```

- Processes all images in `originals`
- Saves to `web` folder
- Moderate quality (good for web)
- Skips any conversion that would increase file size

Example 4: Night Photography

```
img2avif -i night_sky.nef --raw-exposure auto --raw-percentile 0.92 --raw-saturation auto
```

- Auto-exposure targeting 92nd percentile (preserves star highlights)
- Auto-saturation compensates for brightening
- Result: Stars remain visible, sky brightens naturally

Example 5: Batch with Deletion

```
img2avif -r ./raw-imports --delete --quality 85 --keep-metadata
```

- Recursively processes all images in `raw-imports`
- Deletes originals after successful conversion
- High quality for archival
- Preserves EXIF metadata

Example 6: Correcting Wrong Orientation

```
img2avif -i sideways.jpg --rotate 90
```

Forces 90° clockwise rotation regardless of EXIF.

Example 7: Lossless PNG to AVIF

```
img2avif -i diagram.png --lossless --color-space rgb
```

Mathematically lossless conversion preserving all details - ideal for diagrams, screenshots, and graphics with text.

Example 8: Smallest Possible Files

```
img2avif -d ./for-email -q 60 --color-space yuv420 -s 8
```

Smallest files possible, suitable for email attachments (though quality will be noticeably reduced).

Understanding Output Messages

Normal Operation

```
 img2avif v1.3.0
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Supported formats: JPEG, PNG, BMP, GIF, TIFF, WebP, ICO, RAW...

 Found 24 image(s) in ./photos
Recursive mode: enabled
✓ Converted: DSC_0001.nef -> DSC_0001.avif (850.2 KB, 18.3% of original)
```

Verbose Mode Additional Output

```
 RAW Metadata:
Camera: NIKON CORPORATION NIKON D850
Resolution: 8256 x 5504 pixels
CFA Pattern: RGGB
White Balance: R=2.145, G=1.000, B=1.432
Auto-exposure: 1.24x (target percentile: 95%)
Auto-saturation: 1.05x (based on exposure)
Processing 8256x5504 image...
```

Error Messages

Error	Meaning	Solution
Input path does not exist	File or directory not found	Check the path spelling
Unsupported format	File type not recognized	Check file extension
Failed to decode RAW	RAW file corrupted or unsupported	Try different software, verify file
No valid images found	Directory has no image files	Check file types in directory

Tips and Best Practices

For Photographers

1. **Keep originals** until you verify AVIF quality. Use `--delete` only after inspection.
2. Use **auto-exposure** for batches of similar shots under consistent lighting.
3. Set `--raw-percentile 0.92` for night or high-contrast scenes.
4. **10-bit depth** reduces banding in skies and gradients significantly.

For Web Developers

1. **Quality 75-80** is usually indistinguishable from original.
2. **YUV420** offers the best size/quality tradeoff.
3. **Speed 6-8** is fine for production since you encode once.
4. **AVIF support** is now in all modern browsers.

For Archivists

1. Use `--lossless` or **quality 100** for master copies.
2. Consider `--color-space rgb` for maximum fidelity.
3. **Keep metadata** with `--keep-metadata` for copyright and camera info.
4. **Test a few files** before batch processing entire archives.

General

- **Higher quality doesn't always mean visibly better** - test at 80, 85, 90 to find your sweet spot.
 - **RAW files take longer** due to demosaicing (but results are worth it).
 - **Use verbose mode `-v`** the first time you process a new camera's RAW files to verify settings.
-

Supported Input Formats

Standard Formats

JPEG, PNG, BMP, GIF, TIFF, WebP, ICO

RAW Formats (over 50 types)

Nikon (.nef, .nrw), Canon (.cr2, .cr3, .crw), Sony (.arw, .srf, .sr2), Fujifilm (.raf), Olympus (.orf), Panasonic (.rw2), Pentax (.pef), Sigma (.x3f), Adobe (.dng), and many more including .raw, .rw1, .mrw, .dcr, .kdc, .3fr, .ari, .bay, .cap, .data, .dcs, .drf, .erf, .fff, .iiq, .mef, .mos, .mdc, .obm, .ptx, .pxn, .qtk, .rdc, .srw, .sti, .rwz

Version Information

This manual covers **img2avif version 1.2.0**

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